

```

#include <stdio.h>
#include <unistd.h>

int main()
{
    printf("Before exec()\n");

    execl("/bin/ls", "ls", "-l", NULL);

    printf("This line will not execute if exec() succeeds.\n");

    return 0;
}

```

```

sravya@Sravya1955:~/os$ nano os1.1.1.c
sravya@Sravya1955:~/os$ gcc os1.1.1.c -o os
sravya@Sravya1955:~/os$ ./os
Before exec()
total 116
drwxr-xr-x  8 sravya sravya  4096 Aug  1 03:46 MyProject
-rw-r--r--  1 sravya sravya    70 Aug  5 03:48 destination.txt
drwxr-xr-x  2 sravya sravya  4096 Jul 31 09:34 exec
-rwxr-xr-x  1 sravya sravya 16224 Aug  5 03:48 filecopy
-rw-r--r--  1 sravya sravya   681 Aug  5 03:32 filecopy.c
-rwxr-xr-x  1 sravya sravya 16008 Aug  5 04:00 os
-rw-r--r--  1 sravya sravya    93 Jul 29 03:11 os.c
-rw-r--r--  1 sravya sravya   207 Aug  5 04:00 os1.1.1.c
-rw-r--r--  1 sravya sravya   872 Aug  5 03:22 os1.1.c
-rw-r--r--  1 sravya sravya   713 Aug  5 03:27 os1.2.c
-rw-r--r--  1 sravya sravya   680 Aug  5 03:34 os1.3.c
-rw-r--r--  1 sravya sravya   508 Aug  5 03:53 os1.4.c
-rw-r--r--  1 sravya sravya    89 Jul 29 03:17 os1.c
-rw-r--r--  1 sravya sravya     0 Jul 29 03:16 os1.xc
-rw-r--r--  1 sravya sravya   203 Aug  1 02:49 os14.c
-rw-r--r--  1 sravya sravya   243 Aug  1 02:50 os15.c
-rw-r--r--  1 sravya sravya   154 Jul 29 03:21 os2.c
-rw-r--r--  1 sravya sravya   279 Jul 29 03:23 os3.c
-rw-r--r--  1 sravya sravya   214 Jul 29 03:26 os4.c
-rw-r--r--  1 sravya sravya   116 Jul 29 03:26 os5.c
-rw-r--r--  1 sravya sravya   220 Jul 29 03:29 os6.c
-rw-r--r--  1 sravya sravya   132 Jul 29 03:30 os7.c
-rw-r--r--  1 sravya sravya   623 Jul 29 03:38 os8.c
-rw-r--r--  1 sravya sravya    70 Aug  5 03:30 source.txt
-rw-r--r--  1 sravya sravya     0 Jul 28 06:58 test.txt

```

2.

```

#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <stdlib.h>

int main()
{
    pid_t pid = fork();

    if(pid == 0)
    {
        printf("Child Process (PID=%d)\n", getpid());
        sleep(2);
        printf("Child Finished\n");
        exit(0);
    }
    else
    {
        printf("Parent Waiting...\n");
        wait(NULL);
        printf("Parent Continues after Child.\n");
    }

    return 0;
}

```

```

sravya@Sravya1955:~/os$ nano os1.1.2.c
sravya@Sravya1955:~/os$ gcc os1.1.2.c -o os
sravya@Sravya1955:~/os$ ./os
Parent Waiting...
Child Process (PID=2967)
Child Finished
Parent Continues after Child.

```

3.

```
#include <stdio.h>
#include <unistd.h>

int main()
{
    fork();

    fork();

    printf("PID = %d PPID = %d\n", getpid(), getppid());

    sleep(10);

    return 0;
}
```

```
sravya@Sravya1955:~/os$ nano os1.1.3.c
sravya@Sravya1955:~/os$ gcc os1.1.3.c -o os
sravya@Sravya1955:~/os$ ./os
PID = 3015 PPID = 3014
PID = 3017 PPID = 3015
PID = 3016 PPID = 3014
PID = 3014 PPID = 1515
```

```
srayya@Srayya1955:~/os$ strace ls
execve("/usr/bin/ls", ["ls"], 0x7fff27a48050 /* 26 vars */) = 0
brk(NULL)                               = 0x5d17c5a91000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x733ec8ae0000
access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=19763, ...}) = 0
mmap(NULL, 19763, PROT_READ, MAP_PRIVATE, 3, 0) = 0x733ec8ae6000
close(3)                                = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0... ", 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=174472, ...}) = 0
mmap(NULL, 181960, PROT_READ, MAP_PRIVATE, MAP_DENYWRITE, 3, 0) = 0x733ec8ab9000
mmap(0x733ec8abf000, 118784, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x6000) = 0x733ec8abf000
mmap(0x733ec8adc000, 24576, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x23000) = 0x733ec8adc000
mmap(0x733ec8ae2000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x29000) = 0x733ec8ae2000
mmap(0x733ec8ae4000, 5832, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x733ec8ae4000
close(3)                                = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\220\243\2\0\0\0\0\0... ", 832) = 832
pread64(3, "\6\0\0\0\4\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0... ", 784, 64) = 784
fstat(3, {st_mode=S_IFREG|0755, st_size=2125328, ...}) = 0
pread64(3, "\6\0\0\0\4\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0... ", 784, 64) = 784
mmap(NULL, 2170256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x733ec8800000
mmap(0x733ec8828000, 1695632, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x733ec8828000
mmap(0x733ec89b0000, 323584, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1b0000) = 0x733ec89b0000
mmap(0x733ec89ff000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1fe000) = 0x733ec89ff000
mmap(0x733ec8a05000, 52624, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x733ec8a05000
close(3)                                = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libpcr-2.8.so.0", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0... ", 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=625344, ...}) = 0
mmap(NULL, 627472, PROT_READ, MAP_PRIVATE, MAP_DENYWRITE, 3, 0) = 0x733ec8a1f000
mmap(0x733ec8a21000, 450560, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2000) = 0x733ec8a21000
```

```

sravya@Sravya1955:~/os$ nano os1.1.5.c
sravya@Sravya1955:~/os$ gcc os1.1.5.c -o os
sravya@Sravya1955:~/os$ ./os
MiniShell>
Child PID : 3647
Invalid Command
Parent PID : 3638
MiniShell>

```